National University of Computer & Emerging Sciences

Karachi Campus

Event & Venue Booking Management System

Software Engineering

Section: 6B

Group Members:

23k-0633 Khawaja Hassaan Tariq (Group Leader)

23k-0557 Syed Rayyan Amir

1. Overview

This project aims to develop a comprehensive **Event & Venue Booking Management System** to streamline the process of finding, booking, and managing event spaces.

Motivation: The primary motivation is to solve the chaotic and fragmented nature of venue booking, where double-bookings and lack of transparency are common issues.

Customer: The target customers are venue owners (who need management tools) and event planners or individuals (who need to book spaces).

Deliverables: The project will deliver a web-based platform featuring real-time availability calendars, secure booking workflows, and QR-based guest entry systems.

Dependencies: The system depends on external payment gateways (like Stripe/PayPal) and email notification services to function correctly.

2. Project Detail

2.1 Problem or Challenge

Local Challenge: In our current local context, booking a venue for an event (wedding, corporate meeting, or party) is a manual and error-prone process. Venue owners rely on paper registers or basic spreadsheets, leading to scheduling conflicts and lost revenue. Customers often have to physically visit multiple locations just to check availability and pricing, which is inefficient and time-consuming. There is no centralized digital solution that handles the entire lifecycle from booking to guest entry.

2.2 Project Vision

Vision Statement: Our vision is to digitize the local event industry by creating a centralized marketplace where venue booking is as easy as booking a hotel room.
Description: We aim to eliminate administrative headaches for venue owners by automating approvals and payments. For users, we envision a "one-stop-shop" where they can secure a venue, manage their budget, and handle guest entry via QR codes in a single polished interface. The ultimate goal is to increase transparency and trust between facility owners and the general public.

2.3 Project Feasibility Study

2.3.1 Technical Feasibility

The project is technically feasible as it utilizes standard web development technologies. We will use a robust database (MySQL/PostgreSQL) to handle relational data like bookings and cancellations. The integration of QR code generation libraries and email APIs is well-documented and supported in modern frameworks.

2.3.2 Operational Feasibility

The system is designed to be user-friendly, ensuring that non-technical venue staff can easily operate the dashboard. By automating manual tasks like checking dates and sending confirmations, the system reduces the operational burden on venue owners, making adoption highly likely.

2.3.3 Economic Feasibility

The project is economically viable as it relies on open-source development tools, minimizing initial costs. The potential for reducing revenue loss from cancellations (via automated policies) and preventing double-booking errors provides significant financial value to venue owners.

2.4 Project Goals

- **Functional Goals:** To provide a searchable venue listing, a real-time availability calendar, and a secure payment tracking system.
- **Technological Goals:** To implement a secure approval workflow and integrate a QR-based entry system for real-time guest validation.
- **Quality Goals:** To ensure a polished, responsive UI that loads quickly and offers an intuitive user experience.
- **Organizational Goals:** To develop competence in full-stack web development and database management within the team.
- **Constraints:** The system must comply with local data privacy regulations regarding the handling of user contact lists and payment information.

2.5 Project Abstract

The **Event & Venue Booking Management System** is a web application designed to modernize the traditional venue reservation process. It addresses the lack of transparency and efficiency in the current market by providing a dual-interface platform for venue owners and event planners. Key features include a dynamic availability calendar to prevent double bookings, a comprehensive approval workflow for vetting requests, and a QR-code-based guest management system for seamless event entry. The system also includes financial tools such as budget tracking, payment monitoring, and automated cancellation handling. By centralizing these functions, the project aims to reduce administrative overhead and provide a polished, professional experience for all stakeholders.

2.6 Project Scope

In Scope:

- User registration and role management (Admin, Venue Owner, User).
- Venue listing with photos, amenities, and pricing.
- Booking lifecycle management (Request -> Approval -> Payment).
- Guest list management and QR code generation.
- Admin analytics and revenue reporting.

Out of Scope:

- In-house catering services (platform only connects vendors).
- Live streaming of events.
- Hardware procurement for entry scanning (software solution only).

2.7 Project Team

Members	Availability	Comment
Hassaan Tariq	15 hrs/week	Frontend Dev & UI Design
Syed Rayyan Amir	15 hrs/week	Backend Dev & Database

2.8 Existing & Related Work

Application	Features	Limitations	Uniqueness of Our Project
Cvent	Enterprise event management	Expensive and complex for small local venues.	Focused on local SME usability & affordability.

Application	Features	Limitations	Uniqueness of Our Project
Eventbrite	Ticketing focus	Focuses on tickets, not venue reservation/management.	Combines Venue Booking + Guest Entry.
Google Calendar	Scheduling	No payment, vendor, or guest management.	Specialized for Event workflows & Finance.

2.9 Proposed Solution & Architecture

Solution: We propose a three-tier web architecture. The **Presentation Layer** will be a responsive web interface for users to browse and book. The **Application Layer** will handle logic like conflict detection (availability), approval workflows, and QR generation. The **Data Layer** will store user profiles, booking history, and transaction logs.

2.10 Use Case Model

Actors:

- **Guest:** Receives invitation, presents QR code for entry.
- **Event Planner (User):** Searches venues, sends booking requests, manages guest lists, pays.
- **Venue Owner:** Manages listings, approves/rejects requests, views earnings.
- **Admin:** Oversees platform activity, manages disputes.

2.11 Feature List

1. **Venue Listing:** Detailed pages with images, capacity, and amenities.
2. **Availability Calendar:** Visual calendar showing booked vs. available dates.
3. **Booking Request System:** Users request dates; prevents instant booking conflicts.
4. **Approval Workflow:** Owners manually approve or reject requests.
5. **Payment Tracking:** Log deposits, partial payments, and final settlements.
6. **Cancellation Policy Handling:** Auto-calculate refunds based on set rules.
7. **Guest List Management:** Add/Edit guests and track RSVP status.

8. **QR-Based Entry System:** Generate unique QR codes for every guest.
9. **Entry Scanning:** Validator tool to scan QRs at the door.
10. **Vendor Management:** Link external vendors (caterers/decorators) to a booking.
11. **Resource Allocation:** Assign specific tables/chairs to a booking.
12. **Budget Tracking:** Monitor total spend vs. planned budget.
13. **Event Scheduling:** Timeline builder for the event day agenda.
14. **Email Notifications:** Auto-alerts for status changes (Approved/Rejected).
15. **Admin Analytics:** Dashboard showing total bookings and revenue.
16. **Waitlist Mechanism:** Users can join a waitlist for booked dates.
17. **Dispute Resolution:** Ticket system for refund/service issues.
18. **Invoice Generation:** Auto-generate PDF invoices for bookings.

2.12 Software Development Methodology

We will utilize the **Agile Scrum Methodology**. This iterative approach allows us to develop the system in 2-week sprints.

- **Sprint 1:** Database design & Authentication.
- **Sprint 2:** Venue Management & Search.
- **Sprint 3:** Booking Workflow & Payments.
- **Sprint 4:** Guest Management & QR System. This ensures we can test features early and refine the "polished UI" requirement continuously.

2.13 Proposed Infrastructure Platform and Tools

- **Frontend:** React.js / HTML5 / CSS3 (for polished UI).
- **Backend:** Node.js or PHP (Laravel).
- **Database:** MySQL or MongoDB.
- **Design Tools:** Figma (for UI Mockups).
- **Diagramming:** Draw.io.
- **Version Control:** Git/GitHub.

2.14 Project Timeline

- **Weeks 1-3:** Requirement Gathering & SRS.
- **Weeks 4-6:** Design (UI/UX & DB Schema).
- **Weeks 7-10:** Backend Development & API integration.
- **Weeks 11-13:** Frontend Implementation.
- **Weeks 14-15:** Testing & Bug Fixes.
- **Week 16:** Final Presentation.